

Aditya Khichi

adityakhichi763@gmail.com | +91-9039420396
[LinkedIn](#) [GitHub](#)

I am a software engineer with 2+ years of Python backend experience building FastAPI services, data pipelines, and production RAG/LLM workflows on AWS. I have worked with PostgreSQL, Redis, Celery, Docker, and CI/CD, and deployed LLM-powered features with Azure OpenAI. I enjoy owning end-to-end implementations and collaborating closely with stakeholders to turn requirements into reliable systems.

Skills

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|---------------------------------------|---------------|---------------------|-------------|
| • Python | • APIs & REST | • AWS | • LangChain |
| • FastAPI | • PostgreSQL | • Docker | • LangGraph |
| • Backend development | • Redis | • LLM orchestration | • NoSQL |
| • Microservices & distributed systems | • Celery | • RAG | • Kafka |

Employment History

Software Engineer I • Accrete.ai Dec 2023 – Feb 2026

- Developed Python-based backend services powering an LLM chat assistant with long-term memory, real-time contextual retrieval, and Redis-backed session caching for reliable multi-turn interactions.
- Designed and implemented a PostgreSQL + pgvector retrieval pipeline with LangChain, defining low-level backend architecture (chunking strategy, semantic retrieval, ranking) to improve answer relevance.
- Engineered autonomous AI agent workflows in LangGraph with prompt evolution and adaptive response generation, reducing multi-step pipeline latency by 40%.
- Optimized Azure OpenAI integration by tuning embeddings and context-window management, cutting LLM inference latency from 3 minutes to 30 seconds while maintaining output quality.
- Collaborated with cross-functional teammates on backend design and code reviews, improving code quality, observability, and reliability for user-facing AI systems.

Backend Engineering Intern • Accrete.ai Jul 2023 – Nov 2023

- Built FastAPI endpoints in Python for large-scale document processing using Celery workers and Redis as the message broker, increasing the reliability of internal data pipelines.
- Implemented Redis caching for frequently accessed document metadata to reduce redundant PostgreSQL reads and improve API response latency.
- Wrote pytest unit tests and participated in code reviews to strengthen code quality and deepen understanding of Python async patterns and backend design principles.

Developer • Personal project Mar 2026

- ContextMindLM | Built a full-stack distributed RAG platform using Python on AWS ECS Fargate with CloudFront, enabling reliable knowledge retrieval across multiple content sources.
- Developed a document ingestion pipeline with asynchronous processing via Celery and Redis task queues and multi-file uploads via pre-signed S3 URLs for scalable data onboarding.
- Implemented three retrieval modes—semantic, hybrid (BM25 + vector), and multi-query with configurable chunking, thresholds, and reranking to improve retrieval quality and control.
- Designed a LangGraph-based multi-agent orchestration layer to route queries across retrieval, web search, and tool-calling agents, adding structured logging, I/O guardrails, and schema validation for better observability and robustness.
- Project links: <https://contextmindlm.vercel.app/>, <https://github.com/AdityaKhichi/ContextMindLM-server>

Developer • Personal project May 2026

- Shipyard | Built a self-hosted deployment platform that cloned Git repositories, containerized builds on AWS ECS, and served output via S3 for streamlined application deployment.
- Implemented real-time build log streaming using Valkey pub/sub with Socket.IO, removing the need for polling and providing immediate feedback during deployments.
- Designed a reverse-proxy layer that dynamically resolved subdomains to the correct S3 bucket, enabling isolated deployments without manual DNS configuration and simplifying multi-tenant hosting.
- Project link: <https://github.com/AdityaKhichi/Shipyard>

Education

Bachelor of Engineering in Computer Science and Engineering, Mahakal Institute of Technology, Ujjain